Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/26/2025
PI Name:	Szewczyk, Nathaniel Ph.D.		
Project Title:	The Effect of Different Genetic Mutations and Pharmacologic Interventions on Transcriptional Responses to Spaceflight in C. elegans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	szewczyk@ohio.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	740.593.9169
Organization Name:	Ohio University		
PI Address 1:	1 Ohio University		
PI Address 2:			
PI Web Page:			
City:	Athens	**State:**	OH
Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	01/01/2024	**End Date:**	06/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2025 per F. Hernandez/NASA ARC. Original end date was 12/31/2024 (Ed., 1/24/25).		
Key Personnel Changes/Previous PI:	Note: Due to the passing of the Principal Investigator (PI), this is the final report for this investigation (Ed., 9/29/25).		
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K0433		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Long-term human deep space exploration is a key goal of NASA but remains hindered by obscure (mechanistic) knowledge of how spaceflight maladaptation can be overcome. The tiny worm Caenorhabditis elegans represents a primary model for better understanding causes of, and countermeasures against, health decline in space, with strong forward-translation potential into people. Accordingly, a recent spaceflight experiment led by the UK Space Agency (Molecular Muscle Experiment 2 / MME2) flew various mutated or pharmacologically treated worm strains aboard the International Space Station with emphasis on ameliorating spaceflight dysregulation related to neuromuscular, musculoskeletal, and/or mitochondrial signaling (key systems negatively impacted by spaceflight in, and thus highly relevant to, the human system). NASA's GeneLab will generate GeneLab Data Set (GLDS) RNA sequencing (RNA-seq) data from these samples, which the PI proposes to analyse as a GeneLab Analytical Investigation. The central objective is to establish spaceflight transcriptomic profiles for the various mutated/treated strains (using standard transcriptomic methods, including differential expression and network analysis, pathway enrichment, hub gene identification, etc.) and compare them with 'normal' spaceflight responses (i.e., in wild-type worms from this mission plus past flights) to define common/unique effects toward ameliorating spaceflight gene dysregulation. By harnessing the power of omics and the GLDS, this work will therefore establish control mechanisms and efficacy of pharmacological interventions to positively alter gene expression changes during spaceflight, with findings having strong potential to be directly translatable to understand and, ultimately, counteract human health risks for future longer-term deep space exploration. As such, this work will beneficially impact NASA's Space Biology Program and its long-term goals by helping to (i) identify the underlying mechanisms and networks that govern biological processes in space, (ii) develop mechanistic understanding to support human health in space, and (iii) promote open science through the GLDS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New bioinformatics approaches to health data.
Task Progress:	Both objectives have been met. Wild-type worms display alterations in gene expression consistent with past flights. These changes include alterations in genes associated with telomeres, mitochondrial metabolism, muscle function, and stress response. None of the drugs or mutations tested reverse the impact of spaceflight on the alterations observed in wild-type . However, several of the drugs and mutations appear to ameliorate the transcriptional responses vs. wild-type. Additionally, many of the interventions induce transcriptional changes not observed in the wild-type, suggesting additional effects of the interventions beyond modulating the response to spaceflight.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Roberts BM, Deane CS, Szewczyk NJ, Fajardo VA, Maden-Wilkinson T, Bagley JR. "Musculoskeletal responses to spaceflight: Mechanisms, countermeasures, and key gaps." Am J Physiol Cell Physiol. 2025 Dec 1;329(6):C1985-C1993. Review. https://doi.org/10.1152/ajpcell.00649.2025 ; PMID: 41217000 , Dec-2025